



Hypothesis Testing, Confidence Intervals and P-values

Making evidence-based decisions from samples

Simple explanations • worked examples • practical use cases



1. Hypothesis testing

Hypothesis testing checks whether sample evidence is strong enough to challenge an existing assumption.

Waiting-time example

A bank claims that average customer waiting time is 10 minutes.

H0: mean = 10 minutes H1: mean is not 10 minutes

- H0, the null hypothesis, represents no difference or the existing claim.
- H1, the alternative hypothesis, represents the difference being investigated.
- Reject H0 when evidence is sufficiently strong; otherwise fail to reject H0.

2. P-values

The p-value measures how unusual the observed result, or something more extreme, would be if H_0 were true.

If $p \leq 0.05$: reject H_0 If $p > 0.05$: fail to reject H_0

If $p = 0.03$, the result would be fairly unusual under H_0 . It does not mean there is a 3% probability that H_0 is true.

A p-value does not tell us

- The probability that H_0 is true.
- The probability that the result happened by chance.
- Whether the effect is large or useful in practice.
- Whether the data collection and test assumptions were valid.

3. Confidence intervals

A confidence interval gives a range of population values compatible with the sample and method.

Suppose estimated waiting time is 12 minutes with a 95% confidence interval of 11.2 to 12.8 minutes.

Across many repeated samples, about 95% of intervals constructed by this method would contain the true population mean.

- It is not a range containing 95% of individual waiting times.
- It is not a guarantee that this particular interval contains the true value.
- Narrow intervals indicate greater precision; wide intervals indicate greater uncertainty.

4. Combined interpretation and banking use case

For a two-sided test at $\alpha = 0.05$, if the null value lies outside the 95% confidence interval, H_0 is rejected. Here, 10 is outside 11.2 to 12.8, so the data indicates a difference.

Question	Statistical method	Practical implication
Did waiting time change?	Hypothesis test and p-value	Significance is not operational importance
What values are plausible?	Confidence interval	Check width and assumptions
How large is the change?	Effect size	Interpret in business context
Is the result dependable?	Replication and data-quality checks	One test is not final proof

Report the estimate, confidence interval, p-value, sample size and practical importance together.